A Vision-First Multi-Modal Approach for Hand Gesture Recognition

Interactive Puzzle Game Control via Skeleton & Appearance Methods

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Computer Vision Course

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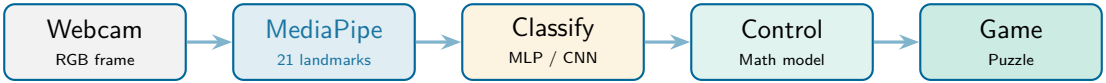
Research Questions



Q1: Multi-Modal

Does fusing skeleton + appearance features improve recognition?

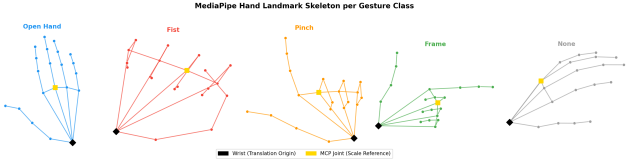
Q2: Model Size

What is the lightest model sufficient for reliable control?

Q3: Interaction

How to bridge classification to jitter-free game control?

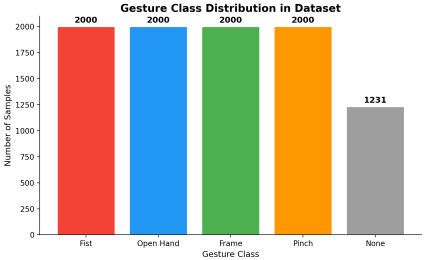
Dataset: HaGRID v2 — 5 Gesture Classes



MediaPipe skeleton landmarks per gesture class

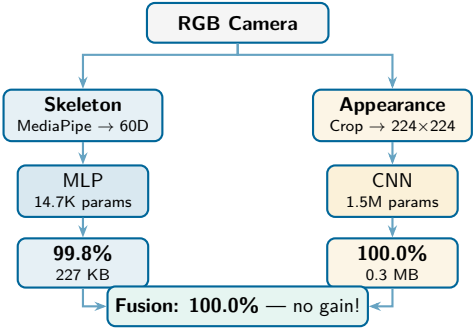
HaGRID v2 Subset:

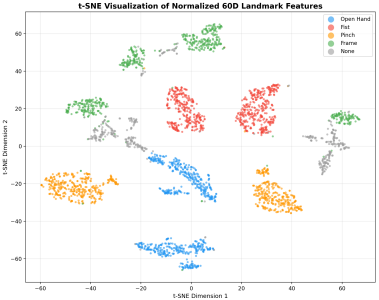
- **23,850** samples
- **13,250** unique subjects
- 5 classes: Pinch, Fist, Open Hand, Frame, None



3:1 imbalance (Frame/None vs others)

Two Feature Streams from One Camera





t-SNE of 60D skeleton features
Clusters already well-separated

→ skeleton alone may suffice

Evaluation: Person-aware

Group 5-Fold CV, 13,250 subjects

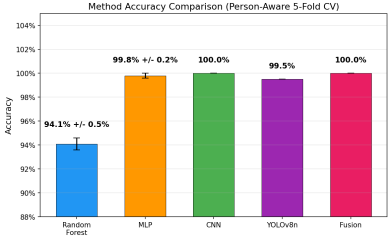
Six Methods: Simple → Complex

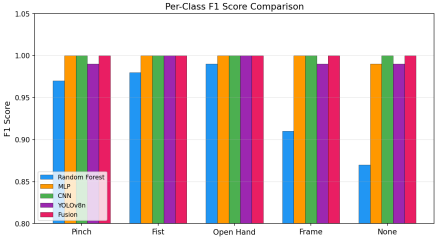
Method	Modality	Acc	Params	Size
Random Forest	Skeleton	94.1%	∼100K	8.6 MB
MLP	**Skeleton**	**99.8%**	**14.7K**	**227 KB**
CNN	Appearance	100.0%	1.5M	0.3 MB
YOLOv8n	Detection	99.5%†	3.0M	6.2 MB
Fusion	**Both**	**100.0%**	**—**	**0.5 MB**

†mAP@50. Heuristic baseline omitted (1%).

The progression:

1. Classical ML: 94% (close but not enough)
2. Neural nets saturate (≥99.5%)
3. Fusion adds nothing over MLP alone

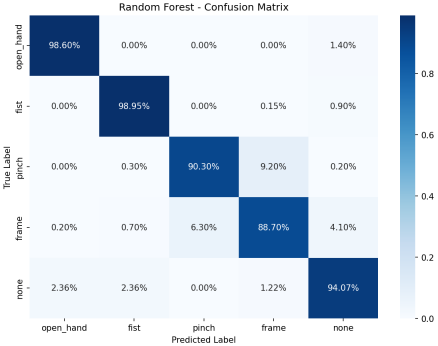




Weakest: "None" (heterogeneous negatives)

Where Do Errors Happen?

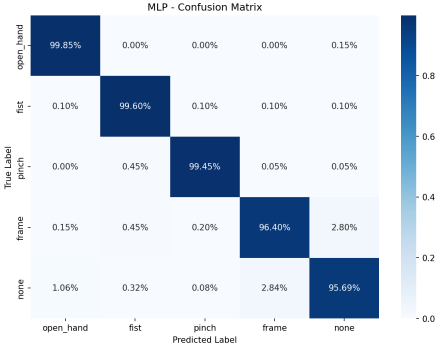
Random Forest (94.1%)



Frame ↔ None confusion

24 engineered features can't resolve it

MLP (99.8%)



Full 60D landmarks resolve ambiguity

Only 0.2% residual error on "None"

YOLOv8n: End-to-End Detection Baseline

Why include detection?

- Classification assumes hand is already found
- YOLO detects + classifies in one pass
- Fallback if MediaPipe fails

Metric	Score
mAP@50	99.5%
mAP@50:95	90.7%
Parameters	3.0M
Model size	6.2 MB

Trade-off for deployment:

- MediaPipe already provides hand detection
- YOLO is **27× larger** than MLP
- Adds localization overhead we don't need

Verdict

MLP on MediaPipe landmarks provides the best accuracy-to-size ratio for our browser deployment.

Key Finding: The Ceiling Effect

Both modalities **independently** saturate accuracy.
Fusion adds **zero measurable benefit**.

Why does this happen?

1. 5 gestures are **geometrically distinct** (fist $\neq$ open hand $\neq$ pinch)
2. Person-aware CV: no identity leakage
3. 13,250 subjects $\rightarrow$ robust generalization

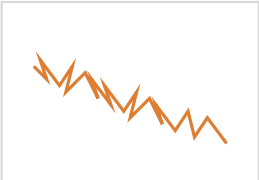
Implication:

1. **Task difficulty** drives accuracy, not model capacity
2. For simple vocabularies: a tiny MLP suffices
3. Fusion helps at harder tasks (full HaGRID: 18 classes, SOTA 95.5%)

Answer to Q1 & Q2: Deploy a 227 KB MLP on skeleton. Fusion unnecessary.

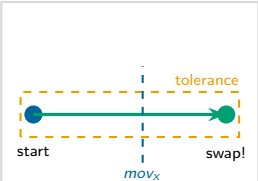
The Real Problem: Classification $\neq$ Control

Without Interaction Model



Hand jitter $\rightarrow$ erratic cursor
False swipes, dropped tiles

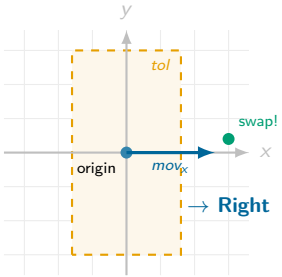
With Interaction Model



Thresholds filter jitter
Only intentional swipes trigger

Inspired by: Huda et al. 2025 — same problem in wheelchair control.
98.14% recognition accuracy, but needed a mathematical threshold model for reliable steering.

Mathematical Interaction Model (adapted from Huda et al. 2025)



Huda: wheelchair steering

Ours: puzzle tile swap

Let $\Delta x = x_{cur} - x_{init}$, $\Delta y = y_{cur} - y_{init}$

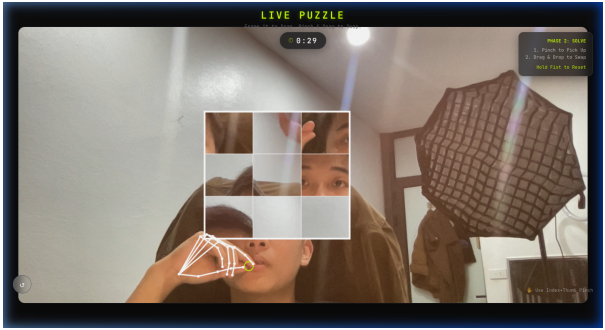
Game Action	Condition
Swap Right	$\Delta x > mov_x$ AND $\lvert\Delta y\rvert < tol$
Swap Left	$\Delta x < -mov_x$ AND $\lvert\Delta y\rvert < tol$
Swap Up	$\Delta y < -mov_y$ AND $\lvert\Delta x\rvert < tol$
Swap Down	$\Delta y > mov_y$ AND $\lvert\Delta x\rvert < tol$

$$mov_{x,y} = 0.6 \times \text{tile_size}$$
$$tol = 0.3 \times \text{tile_size}$$

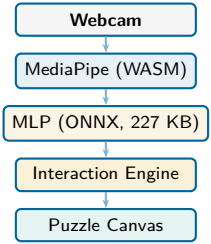
Key design:

- Origin resets after each swap → multi-step swiping
- 3-frame debounce filters transient errors
- Thresholds tied to tile geometry, not pixels

Deployed System



Live puzzle: hand skeleton overlay, timer, gesture instructions



No server. 100% client-side.
30+ FPS on commodity hardware.
227 KB total model size.

Conclusion

Contributions:

1. **Multi-modal finding:** Skeleton (99.8%) and appearance (100%) independently saturate → **ceiling effect**. Fusion unnecessary for 5 coarse gestures.
2. **Interaction model:** Adapted Huda et al.'s directional thresholds (mov_x, mov_y, tol) from wheelchair control → puzzle tile swiping.
3. **Deployed:** 227 KB MLP, 100% client-side, 30+ FPS, no server.

Limitations:

- 5 gestures are trivially separable
- No formal user study
- Fixed thresholds (not per-user)

Future work:

- Full HaGRID (18 classes) where fusion matters
- Adaptive threshold calibration
- Formal usability evaluation

Thank You

Questions & Live Game Demo

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Demo: `https://cv.hieudinh.dev`

Code: `https://github.com/dige04/CVsubject-USTH`

Backup: Frame Gesture & Deployment Differences

Benchmarking vs. Deployed System

Aspect	Benchmark	Deployed
Classifier	MLP (5-class)	MLP (4-class)
Frame detect	MLP	2-hand rule
CNN	Evaluated	Not used
Fusion	Evaluated	Not used

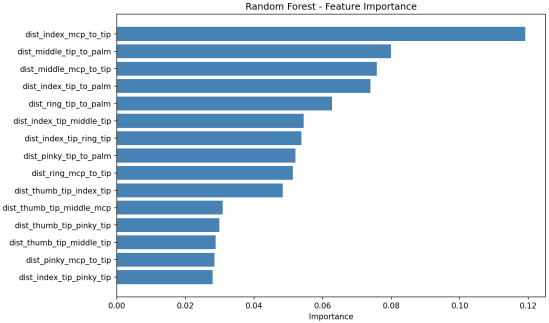
The 99.8% accuracy comes from the 5-class MLP benchmarking study.

Why bypass MLP for "frame"?

- "Frame" = two hands forming rectangle
- MediaPipe reliably detects 2 hands
- Two-hand presence is the **defining property**
- More robust than single-hand crop

Backup: Feature Importance & SOTA Context

RF Feature Importance



A few landmark distances dominate

$\rightarrow$ gestures differ geometrically

HaGRID SOTA (18 classes, 554K):

Method	**Params**	**F1**
MobileNetV3-S	—	57.7
ResNet-18	11M	89.6
ResNet-152	60M	95.5
Ours (5 cls)	**14.7K**	**99.8**

Not directly comparable (5 vs 18),

but shows 5-class ceiling effect.